Upper Respiratory Diseases

NURA 340 Pathophysiology | Lecture Study Guide

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Lecture roadmap: Upper respiratory infections (URIs) → Pharyngitis and tonsillitis → Rhinitis vs. sinusitis → Laryngitis

Building on baseline respiratory physiology, this lecture explores how local infections, immune hyperreactivity, and structural obstructions show up in upper airway disease. Throughout, keep returning to inflammation, immunity, infection, fluid and electrolyte balance, and acid-base equilibrium: they shape every patient's presentation and signs and symptoms.

The recurring mechanism: Pathogen or irritant injures the mucosa → mast cells and macrophages release inflammatory mediators → vasodilation and increased capillary permeability → local edema, mucus, and pain. Most signs and symptoms in this lecture trace back to that chain.

Upper Respiratory Infections (URIs)

Topic	Key points
Incidence	Among the most common acute illnesses worldwide, accounting for millions of healthcare visits a year. Adults average 2–4 episodes per year; children 6–8. Many are common colds, so not everyone needs to see a provider
Transmission	Direct airborne droplets from coughing or sneezing, or indirect contact with contaminated surfaces followed by self-inoculation of the nasal or conjunctival mucosa
Why kids get sick more	Immune systems are not fully built up and have not met as many viral antigens; plus poorer hygiene (not covering coughs, less handwashing, nose picking, hands in mouth, touching surfaces again)
Prevention	Hand hygiene, respiratory etiquette (covering coughs and sneezes), surface disinfection, avoiding close contact with infected people
Complications	Most are self-limiting, but local inflammatory edema can obstruct drainage pathways and lead to secondary bacterial superinfection: acute otitis media (middle ear infection), bacterial rhinosinusitis, abscesses, and eventually spread down to the lower respiratory tract

Fluid, Electrolyte, and Acid-Base Connections

- **Dehydration risk:** exudative secretions, mucosal fluid shifts, fever-induced sweating, faster breathing, and reduced oral intake (not feeling well) can quickly cause fluid volume deficit. This is why pushing fluids is so important.

- **Airflow and CO_2:** severe upper airway mucosal swelling raises airflow resistance, increasing work of breathing and potentially causing CO_2 retention.

Inflammation of the Pharynx and Tonsils

Pharyngitis is inflammation of the mucous membranes of the pharynx. Tonsillitis is inflammation specifically of the palatine tonsils, part of the mucosa-associated lymphoid tissue that guards the entry to the lower respiratory and digestive tracts. These conditions show the immune system's frontline response: pathogens cause mucosal epithelial damage, which activates local mast cells and macrophages. The inflammatory mediators cause microvascular dilation, increased capillary permeability, and tissue swelling. This local inflammatory edema produces the pain, redness, and swelling seen on exam.

Pharyngitis

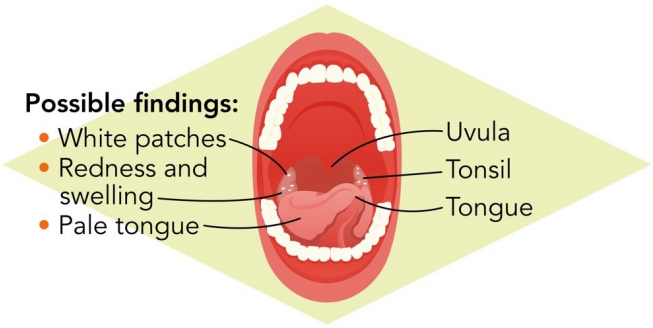
	Details
Viral (most common)	About 70–80% of cases: rhinovirus, adenovirus, seasonal coronavirus, influenza (the common cold)
Bacterial	Primary culprit is group A Streptococcus
Non-infectious	Smoking, air pollution, chemotherapy, GERD, dry air, chronic health problems
Why cause matters	Determines whether treatment is needed; severity usually decides whether patients come in

Pathophysiology

Pathogens adhere to pharyngeal epithelial cells, causing tissue invasion and cell necrosis. This triggers an acute inflammatory cascade: neutrophils and lymphocytes flood the site, forming exudate and inflammation.

Manifestations and why they happen

Manifestation	Mechanism
Sore throat, painful swallowing	Inflammatory mediators sensitize pain nerve endings in the pharyngeal wall; swallowing pushes food or liquid against them
Throat edema and hoarseness	Fluid leaks from the microvasculature into pharyngeal and laryngeal soft tissue, swelling it and altering vocal cord resonance
Nasal congestion and discharge	Vascular engorgement and submucosal gland stimulation cause fluid shifts and mucus hypersecretion

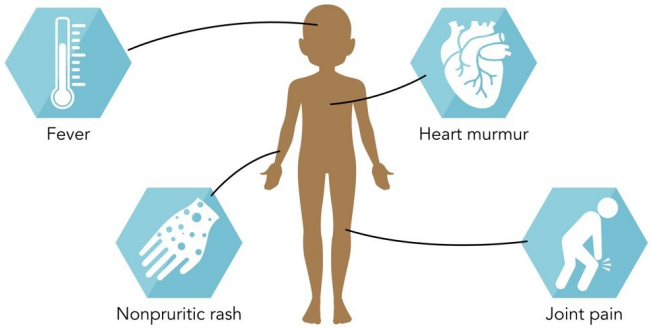


Possible findings in pharyngitis (lecture slide)

Diagnosis

A rapid antigen-detecting test or throat culture when a bacterial (strep) infection is suspected.

Complication to know: Untreated bacterial (strep) pharyngitis can progress to rheumatic fever, a systemic illness with a nonpruritic rash, heart murmur, and joint pain, and cardiac involvement. It is uncommon in the United States because of access to healthcare, but is seen more in low-income areas and among people without easy access to care.



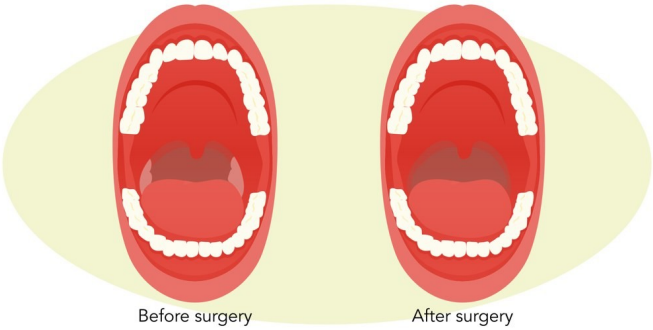
Rheumatic fever signs: fever, heart murmur, nonpruritic rash, joint pain (lecture slide)

Tonsillitis

Inflammation concentrated in the palatine tonsils; can be caused by viral or bacterial pathogens. Pathogen invasion triggers intense leukocyte proliferation in the germinal centers of the tonsillar lymphoid tissue. Infiltration and capillary vasodilation cause tissue hypertrophy (swelling).

Manifestation	Mechanism
Sore, scratchy throat	Prostaglandins and bradykinin sensitize nerve endings, same as pharyngitis; swallowing presses against them
Red, swollen tonsils (hallmark)	Hyperemia reddens the tissue while increased capillary permeability leaks plasma into the interstitial space, causing marked enlargement
Dysphagia	Hypertrophied tonsils narrow the oropharyngeal lumen, a mechanical obstruction. Liquids usually pass, but solid food is harder
Hoarseness	Narrowing alters airflow around the vocal cords, changing acoustic resonance

- **Diagnosis:** throat swabs and rapid antigen detection confirmation.
- **Treatment note:** when the tonsils physically meet in the middle ("kissing tonsils") or cause airway obstruction, surgical removal (tonsillectomy) is needed to open the pharyngeal airway. This is why many people have had their tonsils removed.



Tonsils before and after surgery (lecture slide)

- **Outcome:** most cases are mild without lasting complications. Bacterial infection can lead to systemic infection (rheumatic fever, scarlet fever) or abscesses. Tonsillitis may be a one-time event, recurrent, or frequent.

Rhinitis vs. Sinusitis

Rhinitis is inflammation restricted mainly to the mucous membranes of the nasal cavity. Sinusitis is inflammation of the lining of the paranasal sinuses. Because the nasal mucosa is continuous with the sinus lining, the two often occur together as rhinosinusitis. Both show how epithelial injury triggers local edema and excess fluid, changing airflow and drainage across the upper airway.

Nonallergic Rhinitis

Allergic (IgE-mediated) rhinitis was covered in the immunity lecture. Nonallergic rhinitis is triggered by non-IgE-mediated causes.

- **Triggers:** viral infections, environmental irritants, weather changes, medications, hormonal shifts.
- **Pathophysiology:** autonomic dysregulation or direct irritation causes parasympathetic activation and neurogenic inflammation. Arterial vasodilation engorges the venous sinusoids of the nasal turbinates, and high capillary hydrostatic pressure forces plasma fluid into the interstitium, causing swelling.

Manifestation	Mechanism
Nasal congestion and mucosal edema	Submucosal interstitial fluid accumulates and physically narrows the nasal passages
Rhinorrhea / increased secretions	Mucus hypersecretion and increased capillary permeability
Nasal itching	Mediators stimulate sensory trigeminal nerve fibers in the nasal mucosa

- **Diagnosis:** no specific test. The patient is tested for allergic rhinitis; if negative, it is called nonallergic rhinitis (a rule-out diagnosis).

- **Fluid loss:** profuse rhinorrhea and obligatory mouth breathing (from congestion) increase insensible fluid loss and can worsen dehydration.
- **Outcome:** typically acute and self-resolving, once the virus runs its course, the body adjusts to a medication or weather change, or the medication is stopped. No severe long-term complications.

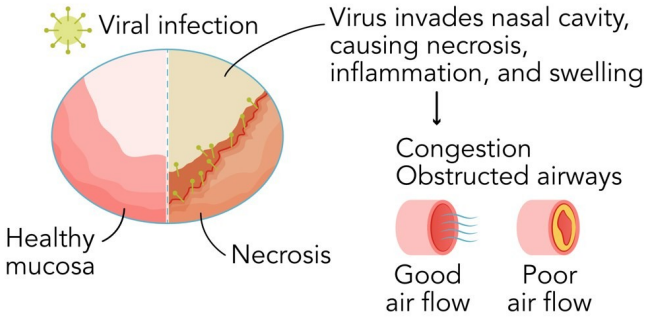
Sinusitis

	Acute	Chronic
Duration	Less than 4 weeks	More than 12 weeks
Cause	Several pathogens; viral most common. But sinusitis usually results from a secondary bacterial infection after the viral illness, not the virus itself	Bacteria and viruses together are the most common causes. Often seen with ear infections, asthma, cystic fibrosis, or AIDS (other inflammatory and immune conditions)
Outcome	Usually resolves on its own without long-term complications	More complications (see below)

Pathophysiology

A step-by-step chain:

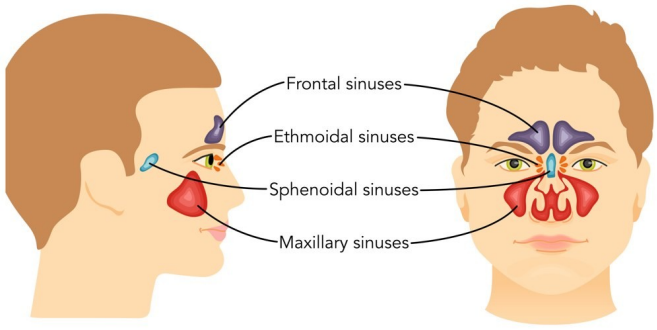
1. **Viral invasion:** cellular necrosis and mucosal edema block the narrowed sinus ostia (the drainage channels).
2. **Sealed cavity:** air and trapped mucus are sealed inside rigid bone cavities. Oxygen is absorbed, creating a negative-pressure vacuum, followed by fluid accumulation.
3. **Stagnant fluid:** it becomes hypoxic and loses normal ciliary clearance, turning into a culture medium for bacterial superinfection.
4. **Airflow:** tissue swelling changes the blood vessels and airways from supporting good airflow to causing poor airflow.



Viral invasion, necrosis, and obstructed airflow (lecture slide)

Manifestations and why they happen

Manifestation	Mechanism
Facial pain and pressure, worse leaning forward	Blocked drainage traps purulent fluid in rigid bone cavities, building positive pressure. Pain is felt below the eyes over the cheekbones. Leaning forward raises hydrostatic pressure in the head and increases tension along the sinus walls. Can be intense in severe infections
Purulent nasal drainage	Static mucus lets bacteria multiply. Neutrophils infiltrate; as white blood cells die, their enzymes turn the fluid thick and yellow or green as it leaks past the obstructed channels. Only a little drains, so there is less of a "head cold" runny nose but very thick mucus
Reduced sense of smell	Edema high in the nasal vault physically blocks odorant molecules from reaching olfactory nerve receptors
Fever	Cytokines enter the systemic circulation during the acute bacterial infection phase



The paranasal sinuses (lecture slide)

Color of drainage: The lecture notes that thick yellow or green drainage suggests true sinus infection, and says green is typically bacterial and yellow viral "from what I've learned." Treat this as a rough cue only: color alone does not reliably tell bacterial from viral, and timing (below) is the more reliable guide.

Diagnosis

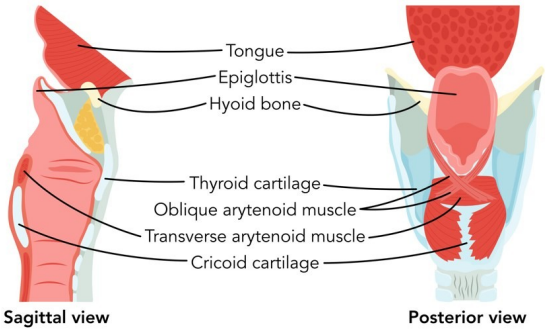
- **Clinical:** more than 10 days of symptoms suggests a bacterial cause; fewer than 10 days suggests viral, which is allowed to run its course.
- **Criteria:** at least two major and two minor manifestations.
- **CT scan:** if unsure, or if chronic sinusitis does not go away (for example, despite antibiotics for a presumed bacterial cause), to visualize mucosal thickening.

Complications

Acute cases rarely have long-term problems. Chronic cases have more, including laryngitis and lacrimal sac (tear duct) inflammation. If untreated and severe (usually bacterial), infection can spread to nearby structures: the eyes, ears, and brain. Spread to the brain is life-threatening, which is the main worry with a sinus infection.

Laryngitis

Inflammation localized in the larynx, typically from viral URIs, vocal strain, or inhaled irritants. Acute mucosal inflammation causes vasodilation and fluid shifts around the vocal cords. The edema thickens and stiffens the cords, disrupting the symmetrical vibration needed to make sound.



Larynx anatomy (lecture slide)

Manifestation	Mechanism
Hoarse voice or loss of voice	Swelling and stiffening of the vocal cords
Dry cough	Inflammatory swelling and lack of lubricating fluid
Sore throat	Cytokines stimulate pain nerve endings

- **Diagnosis:** most people diagnose it themselves (lost voice, dry cough, sore throat). If symptoms persist more than 3 weeks, a scope may be used to look at the larynx.

Quick Review

Condition	Key points
URI	Common; droplet and contact spread; prevention by hygiene; risk of secondary bacterial infection (otitis media, sinusitis); watch for dehydration
Pharyngitis	Mostly viral (70–80%); group A strep is the bacterial cause; rapid antigen test or culture; untreated strep can cause rheumatic fever
Tonsillitis	Red swollen tonsils (hallmark), dysphagia; swabs and rapid antigen; tonsillectomy if tonsils meet or obstruct the airway
Nonallergic rhinitis	Non-IgE triggers; congestion, rhinorrhea, itching; diagnosis by ruling out allergic rhinitis; self-limiting
Sinusitis	Acute < 4 weeks, chronic > 12 weeks; facial pain worse leaning forward, thick yellow/green drainage, reduced smell, fever; > 10 days suggests bacterial; CT if persistent; brain spread is life-threatening
Laryngitis	Hoarseness, dry cough, sore throat; usually viral or vocal strain; scope if > 3 weeks

Notes on Transcript Cleanup

Corrected mis-transcriptions: "non-parodic rash" is nonpruritic rash; "heart murmur from joint pain" is read as heart murmur and joint pain (matching the slide image); "bradykins" is bradykinin; "nasal turbulence" in the rhinitis explanation is read as turbinates; "respiratory adequate" is respiratory etiquette; "lacrimal sac inflammation, which is our tear ducts" is kept as spoken. The line listing complications of chronic sinusitis as "laryngitis" was kept as spoken, though it is not a typical sinusitis complication; check your notes. The yellow vs. green drainage comment is the instructor's own caveated recollection and is flagged in the guide. The slides list "diagnostic criteria" as headings without detail; the criteria shown come from the lecture. The transcript ends with the laryngitis diagnosis, so there was no separate lecture summary.